BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 082 · Linear Search

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a syntax error?

Q2. What does FROM name in SQL?
A. The output columns
B. The source table
C. The filter condition
D. The sort order

Q3. “Encryption prevents all network packets from being intercepted.” Correct this. Explain why.

Q4. A school replaces working laptops each year. “This has no environmental cost.” Correct this. What resources or waste are involved?

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I Do · Read new knowledge

Key vocabulary

Linear search — Method checking array elements in sequence until a match is found.

Target — Value being searched for.

Comparison — Checking an array value against the target.

Index — Zero-based position in an array.

Unsorted — Data not arranged in a particular order.

Not found — Result when no matching element is present.

New learning

Search each position: Linear search starts at the first element, compares it with the target, and moves through elements in order.

Works with unsorted data: The values do not need to be sorted. Linear search can inspect a list containing any arrangement.

Record comparisons and result: Stop once the first target is found (or record it without overwriting), or report not found after every relevant element is checked.

Use array indices: OCR ERL arrays are zero indexed. The fourth element is at index 3; the number of comparisons and array position are not the same.

Worked example

Trace a linear search for target 2 in [8,4,11,2,7]. State the checked values and zero-based position where 2 is found. [4 marks]

1. Begin at index 0 and check 8; it is not 2 [1].

2. At index 1, check 4; still not 2 [1].

3. At index 2, check 11; still not 2 [1].

4. At index 3, value 2 matches. Report position 3 [1].

We Do · Practise with guidance

Guided 1

Trace linear search for 5 in [3,9,5]. State the comparisons and the index found. [3 marks]

Worked method: Check 3 then 9 [1]; check 5 [1]; index found = 2 [1].

Guided 2

e.g. -1 [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A syntax error breaks the language's rules. The affected code cannot be translated or run correctly.

Q2. B — The source table

Q3. Encryption changes plaintext into ciphertext. Someone who intercepts it cannot easily read it without the key.

Q4. Making devices uses energy and raw materials. Disposal adds electronic waste. Consider who benefits and who is harmed.

I Do / We Do pages · Guided practice

Q1. Check 3 then 9 [1]; check 5 [1]; index found = 2 [1].

Q2. Compare 8, 6 and 4 [1]; no element equals 7 [1]; report not found after 3 comparisons [1].

Q3. The first element is 2 [1], so the first matching index is 0 [1].

You Do pages · Code First

```
array names = ["MAR","MDR","PC"]
for i = 0 to 2
    if names[i] == "PC" then
        print(i)
    endif
next i
```

Expected result: 2

You Do pages · Independent questions

Q1. It does not require sorted data [1], so it can search the current unsorted array without sorting it first [1].

Q2. Compare 6 [1], then 3 [1], then 8 [1], then 1 at zero-based index 3 [1].

Q3. Start at index 0 [1], compare 5 then 9 then 2 [1]; at index 3 value 7 matches [1]. Return index 3 [1]. If all entries were checked without a match [1], report not found,

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Search for 7 in [8,6,4]. Explain the outcome and number of comparisons. [3 marks]

Worked method: Compare 8, 6 and 4 [1]; no element equals 7 [1]; report not found after 3 comparisons [1].

Guided 3

Search [2,2,5] for 2 using linear search. Explain the first match position and why. [2 marks]

Worked method: The first element is 2 [1], so the first matching index is 0 [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the comparison that finds "PC" in a register-name array.

```
array names = ["MAR","MDR","PC"]
for i = 0 to 2
    if ... then
        print(i)
    endif
next i
```

write

Question 1

State one advantage of linear search on an unsorted array. [2 marks] Source: Original OCR J277-style question; not an official OCR past-paper original.

Question 2

A list is [6,3,8,1]. Trace linear search for 1, stating each comparison and index. [4 marks] Source: Original OCR J277-style question; not an official OCR past-paper original.

Question 3

A shop has an unsorted array [5,9,2,7,3]. Explain the main steps of linear search for 7, compare successful and unsuccessful outcomes and describe how an index is reported. [6 marks] Source: Original OCR J277-style question; not an official OCR past-paper original.

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Exit Ticket · Answer before checking

Linear search requires an array to be sorted before it can be used.